Heliex: LLM-Powered Empathic Companion for Mental Health Support

Technical Architecture, LoRA Personalization, and Cost Evaluation for a Medium-Sized Startup (2025)

Executive Summary

Heliex is an AI-powered mental-health platform designed to serve as an *empathic companion* with a psychotherapy background.
It bridges the gap between human therapists and AI systems by emulating professional communication styles through LLM-driven personas.
Each registered psychologist provides a brief description of their therapeutic approach and can optionally fine-tune an LLM persona via a lightweight **LoRA (Low-Rank Adaptation)** adapter trained on anonymized dialogue data.

Heliex leverages a **modular, cloud-native architecture** combining safety validation, real-time inference, and dynamic persona loading.
This approach enables both scalability and deep personalization while adhering to clinical safety and data privacy requirements.

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1. System Architecture

Key Components

Layer	Description
Frontend (Client App)	Web/mobile chat interface for users and psychologists.
API Gateway / Proxy	Manages authentication, routing, and rate limiting.
Persona Orchestration Layer	Selects the appropriate persona prompt or loads a LoRA adapter; applies guardrails.
Persona Knowledge Base (KB)	Stores psychologist profiles, summaries, example dialogues, and LoRA adapter checkpoints.
Core LLM Backend	Performs inference using the base model plus optional LoRA adapter injection.
Post-Processor	Applies final safety filters and compliance logging before delivering responses.

This layered structure ensures flexibility, maintainability, and secure separation between user-facing services and sensitive AI components.

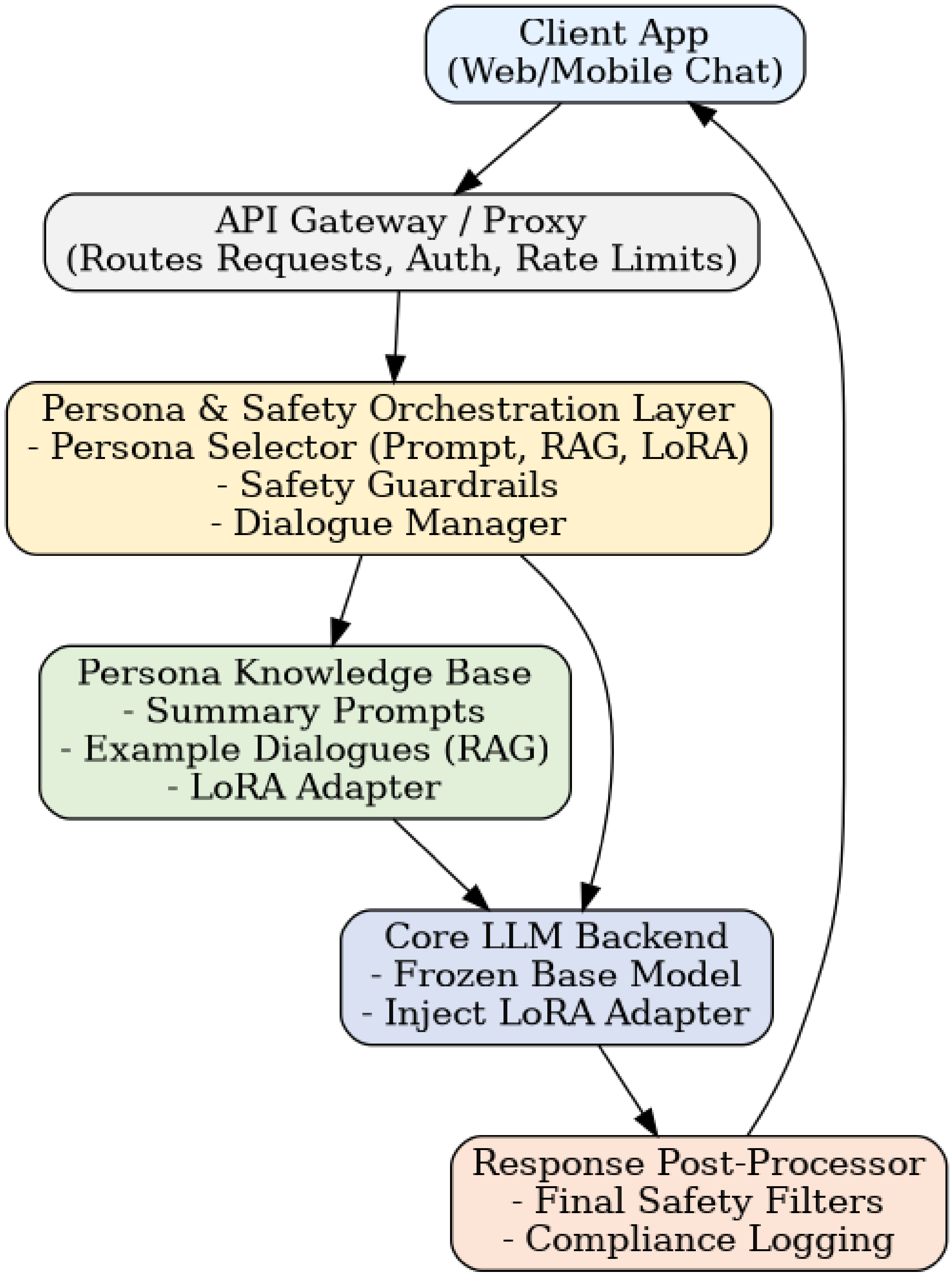

Figure 1: Heliex System Architecture

2. LoRA Adapter Personalization

Low-Rank Adaptation (LoRA) is a fine-tuning method that adds small trainable matrices inside transformer layers, leaving base weights frozen.
Each psychologist's tone and interaction style can be learned efficiently with LoRA adapters that typically require **under 200 MB of storage**.

Heliex LoRA Workflow

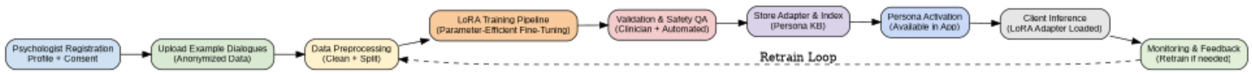

Figure 2: Heliex LoRA Workflow

Advantages - Extremely lightweight and cost-efficient.
- Enables real-time persona switching.
- Preserves base model quality and safety compliance.
- Scalable to hundreds of unique psychologist profiles.

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3. Developer Implementation Deep-Dives

3.1 Data Preprocessing

Objectives: anonymize, standardize, and structure dialogue data for training.

Pipeline: 1. **Anonymization** — NER-based redaction for PII (names, locations, contact info), plus optional human spot checks.
2. **Segmentation** — Split conversations into `(user, therapist)` pairs; filter very short turns.
3. **Normalization** — Unicode normalize, normalize whitespace, strip system artifacts.
4. **Quality Filters** — Deduplicate by hash, remove toxic/off-topic segments unless explicitly permitted for safety tests.
5. **Tokenization** — Use the base model's tokenizer; enforce max sequence length; truncate or sliding-window long sessions.
6. **Formatting** — Store as JSONL or Parquet with fields: `persona_id`, `input`, `output`, `tags`, `timestamp`.

Infra Notes (cloud): - Stateless CPU workers on AWS Fargate / GCP Cloud Run.
- Throughput: ~3 min / 100 convos CPU-side.
- Store clean datasets in object storage (S3/GS) with versioning.

3.2 LoRA Training

Frameworks: Hugging Face `peft`, PyTorch Lightning LoRA.
Base Models: OpenAI fine-tunable endpoint (if available for your plan) or OSS (Mistral 7B, Llama 3 8B).

Typical Hyperparameters: - `r = 8` (rank), `lora_alpha = 16`, `lora_dropout = 0.1`
- `lr = 2e-4`, batch size 16, epochs 3–5
- Optimizer: AdamW (weight decay 0.01)
- Precision: bfloat16/float16

Artifacts: Only adapter weights (e.g., `.safetensors`) — ~50–200 MB.

Cost/Time (AWS A100 spot): - $18–$25 per run (1 hour), depending on dataset size.

Operational Flow (on-demand per psychologist): 1. Psychologist triggers training from dashboard.
2. Preprocessed dataset is fetched (`persona_id`).
3. Launch short-lived GPU job (EKS/SageMaker/Batch).
4. Save adapter + metrics, update Persona KB index.

3.3 Validation & Safety QA

Automated: - Toxicity & Self-harm Classifiers: reject unsafe generations.
- **GuardrailsAI LlmRagEvaluator:** detect hallucinations and factual drift.
- **Jailbreak Detection:** pattern-based + LLM meta-detectors.

Manual: - Clinician Review: sample 10–20 prompts; rate empathy, clarity, appropriateness on Likert scales.
- **Decision:** approve, quarantine, or request retraining.

Release Gate: Adapters not passing automated + manual checks are **not** promoted to production.

3.4 Client Inference

Runtime Steps: 1. User selects a psychologist persona.
2. **Orchestrator**:
- Retrieves persona prompt + (optional) RAG snippets + **LoRA adapter**.
- Assembles a constrained system prompt (ethics & scope).
3. **Core LLM** runs inference with the adapter injected.
4. **Post-Processor** applies final safety filters; logs metadata.

Latency Targets: 2–3 s per message with caching and streaming; LoRA overhead is minimal when adapters are preloaded.

3.5 Monitoring & Retraining

What to Track: - Safety flags, jailbreak attempts, refusal rates.
- Session satisfaction (thumbs up/down), NPS proxy.
- Persona usage distribution; abnormally high/low engagement.

Drift & Retraining Triggers: - Automatic: drift > 15% vs baseline on semantic clusters or sentiment.
- **Manual:** psychologist uploads new examples or requests updates.

Cadence: on-demand per psychologist; common cadence monthly or bi-monthly for active personas.

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4. LLM Selection Strategies

4.1 OpenAI ChatGPT API

- **Pros:** zero infra, enterprise-grade compliance, predictable latency, fast iteration.
- **Cons:** limited weight-level control; LoRA only if/when supported by provider; data residency considerations.

4.2 Open-Source LLMs (Mistral, Llama 3, Falcon, etc.)

- **Pros:** full control of weights/adapters; flexible privacy (VPC); offline/on-prem possible; tunable latency-cost tradeoffs.
- **Cons:** GPU infra + MLOps needed (K8s/EKS, autoscaling, logging, model registry); higher ongoing ops overhead.

Comparison Table

Feature	OpenAI ChatGPT	Open-Source LLM
Hosting	Managed API	Self-hosted (AWS/GCP, EKS/SageMaker/Vertex)
Infra Cost (monthly)	$0 setup	**$3,000–$5,000** (1× A100 + support)
Per-Message Cost	API usage-based (~$1,000–1,500 / 100k msgs)	GPU time + storage; can be cheaper at high volume
Latency	~2 s	~3–5 s (typical), tunable
Customization	Prompts, structured tools	Full LoRA fine-tuning + control
Security/Privacy	External processing, enterprise programs available	Data stays in VPC / on-prem
Compliance	SOC2/HIPAA (provider program)	Your responsibility (audit, logging, access controls)
Maintenance	Low	High (DevOps + ML Ops)

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5. Development Effort & Cost Estimation (USD)

Context: Medium-sized startup, cloud-native, on-demand LoRA per psychologist, 6-month MVP→Prod.

5.1 Dev Effort (One-time, team of 5)

Phase	Scope	Effort	Cost
Architecture & CI/CD	Infra design, pipelines, IaC	6 wks	$25,000

Phase	Scope	Effort	Cost
Frontend & Gateway	Web/mobile chat, auth, routing	8 wks	$35,000
Persona Orchestrator & KB	DB schemas, prompt mgmt, adapter registry	8 wks	$30,000
LoRA Pipeline	Training service + validation + storage	10 wks	$45,000
QA & Security	Safety tests, compliance review	6 wks	$20,000
Deploy & Monitoring	Observability, rollouts	6 wks	$25,000
Total	~6 months	—	**$180,000–$200,000**

5.2 Monthly Ops Costs

Item	ChatGPT API	Open-Source LLM
Inference (100k msgs)	$1,000–1,500	—
LoRA Training (on-demand)	$200–500	$800–1,200
Storage (S3/GS)	$100	$300
Monitoring & Ops	$200	$1,000
Infra (GPU/K8s)	$0	$3,000–5,000
Total / month	**$1,500–2,300**	**$5,000–8,000**

> **Takeaway:** Start on ChatGPT API for speed & cost efficiency; migrate to a hybrid/open-source stack as volume or privacy demands grow.

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6. Strategic Conclusion

Heliex delivers **scalable, ethical, personalized** mental-health conversations by blending a shared LLM backbone with **persona prompts** and optional **LoRA adapters**.
The architecture supports rapid MVP on managed APIs and evolves naturally to a private-cloud/open-source stack when needed — balancing cost, control, and compliance across growth stages.

End of Report
(Heliex 2025 — Confidential Technical Architecture Document)